

Numerical solution of Arrhenius chemical reaction and non-linear radiant of Williamson fluid along a wedge

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Abstract

The foremost importance of this presentation is to explore the Arrhenius activation energy and binary chemical reaction with nonlinear thermal radiation on a Williamson fluid model above a wedge. An aligned magnetic flux and a non-uniform heat source/sink are also included. A study is also on track to analyze the factors such as Brownian diffusion and Thermophoresis. Similarity variables of governing equation are used to reduce into non-dimensional form. The governing equations are derived and solved by using the MATLAB bvp4c method. Corresponding results are plotted in graphs and reviewed specifically. The wall temperature intensifies with the temperature ratio parameter. It is seen that the concentration decreases with the increasing value of chemical reaction parameter and a reverse trend is observed for the enhancement of Eckert number. The temperature and concentration are proportional to Nt and Nb . The mass transfer rate elevates for Chemical reaction rate and shows reverse trend for activation energy.

KEYWORDS: Williamson Fluid, Wedge, nonlinear radiation, non-uniform heat source/sink

INTRODUCTION

Electromagnetic radiation is the core reason for transmitting heat radiation subjected to structure, hotness and behaviour of the material that release or absorb heat, and doesn't needed any means to prorogation. But in industry with a tiny difference in temperature inside the fluid is bothersome. For this reason, scientists included an additional term nonlinear radiation besides radiation. This additional term states the discrimination in temperatures of wall and working fluid. To give an aerial survey a few citations were given in [1-4].

Currently, the thermophysical description of fluid flow over a wedge is exceptional as of its applicability in the fields of chemical industry and engineering. The fluid flow over a wedge has many applications in flows in which proportionality is maintained between free